

FIN — Release 10.23

Second-Generation Atlas

Derived Nadsoliton Objects

Systematic search over C01...C12,
construction of O01...O15, and synthesis of G01...G10

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Central object.

$$E \mapsto \text{Schur}_{V \setminus E}(zI + A), \quad \Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}.$$

Contextual reductions compose, while self-energies are positive operator-valued Stieltjes functions generating exact memory.

Boundary. This report does not export a selector, physical scale, legacy–strict bridge, role transfer, or laboratory validation.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Contents

Confidence convention	1
1 Executive summary	2
2 Generation rather than analogy matching	4
3 Base inventory C01...C12	5
4 Main table: second-generation objects	6
5 Explanatory power and falsification	7
6 Ranking O01...O15	8
7 Theorems generated in the second phase	10
7.1 O01 belongs to the operator Stieltjes class	10
7.2 O03 composes functorially	10
7.3 O02 reconstructs the exact contextual Green function	10
7.4 O05: memory has a chiral response direction	11
7.5 O09: negative informational coupling as channel contraction	11
7.6 O07+O08: exact calibration orbit	12
7.7 O06: heat geometry contracts	12
8 Second-order analogies	13
8.1 Physics	13
8.2 Mathematics	13
8.3 Computer science	14
8.4 Biology	14
8.5 Cybernetics	14
9 Third generation: combining O-objects	15
10 Principal "shadow shapes"	17

10.1 Shadow 1 — the missing object space is Stieltjes space	17
10.2 Shadow 2 — observation and compression both select context	17
10.3 Shadow 3 — "negative information" is often flow into an invisible sector	17
10.4 Shadow 4 — chirality is a tangent, not a point	17
10.5 Shadow 5 — physicalization requires two sections and one process	18
10.6 Shadow 6 — adaptation may act on memory rather than directly on a kernel	18
10.7 Shadow 7 — the bridge defect may be dynamic	18
11 Falsification of the strongest interpretations	19
11.1 Is G02 a new category theory?	19
11.2 Does O05 solve the selector problem?	19
11.3 Is O09 physical entropy?	19
11.4 Does O12 prove a fractal fixed point?	19
11.5 Does O11 prove that FIN is biological?	19
11.6 Does G09 provide a global section?	19
12 Research programs 255–266	20
13 Strategic priority	21
14 Guardrail-compatible boundaries	22
15 Final verdict	23
16 Comparative literature	24
17 Reproducible artifacts	25
Reproduction certificate	26

Confidence convention

- **[Proven]** — an analytic result or an explicit numerical identity within a stated tolerance.
- **[Strong evidence]** — a stable construction whose general theorem or interpretation remains open.
- **[Moderate evidence]** — a typed mechanism whose identifiability or uniqueness remains open.
- **[Speculative]** — a constructive hypothesis intended for falsification.
- **[Refuted]** — a connection rejected in the stated class, usually because no common typed operation exists.

Chapter 1

Executive summary

The second phase of the atlas no longer asks only what the puzzles C01...C12 resemble. It asks:

which new objects can be constructed by composing their operations?

A finite, reproducible search was performed:

Class	Count
individual puzzles	12
all pairs	66
all triples	220
selected couplings of 4–6 puzzles	7
combinations scanned in total	305
rows yielding a typed object or mechanism	109
purely visual combinations rejected	184
second-generation O-objects	15
third-generation G-objects	10

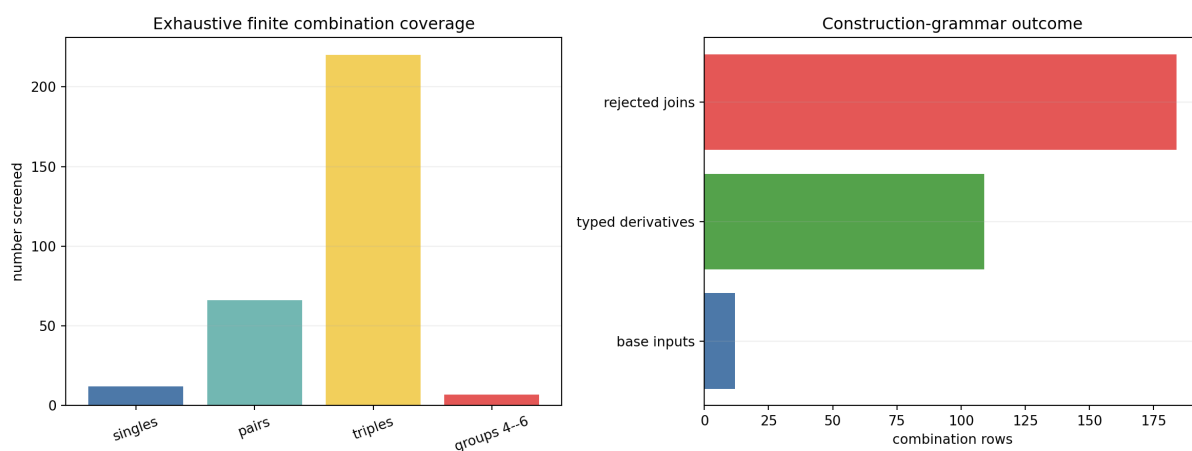


Figure 1.1: Complete coverage of singles, pairs, and triples, together with the output of the explicit construction grammar.

The strongest derived object is not a single matrix but a contextual family:

$$\mathfrak{S}_A : E \mapsto B_E(z) = \text{Schur}_{V \setminus E}(zI + A), \quad z > 0.$$

For every observed context E , the hidden sector generates the self-energy

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}.$$

This family has three exact properties:

1. nested reductions compose exactly by associativity of the Schur complement;
2. $\Sigma_E(z)$ is a positive operator-valued Stieltjes function;
3. its inverse Laplace transform is the memory kernel of the reduced process.

We call the resulting G02 object the **Stieltjes–Schur Context Functor**.

It is **[Proven]** as finite operator algebra and **[Speculative]** as a foundation for physical ontology.

The second particularly promising object is the chiral memory susceptibility

$$\Xi_E(z) = \partial_\theta \Sigma_E(z, \theta)|_{\theta=0}.$$

It is nonzero and odd under reflection, but it is a **receiver** of the inserted twist θ , not a source of orientation. **[Strong evidence]**

The third result clarifies "negative information coupling." Loss of accessible information can be represented without modifying the kernel by the contraction ledger

$$\mathcal{L}_t(p, q) = D(p||q) - D(P_t p || P_t q) \geq 0.$$

This is loss of distinguishability in the accessible channel. It is not automatically destruction of information or thermodynamic energy. **[Proven]**

Chapter 2

Generation rather than analogy matching

Every combination passes through five gates:

1. **Common operation:** can the puzzle operations be composed with compatible input and output types?
2. **New object:** is the output more than a list of constituents?
3. **New space:** does the object live in a recognizable space of operators, functionals, measures, or functors?
4. **Observable and dynamics:** does the construction provide a testable number, record, or evolution?
5. **Falsification gate:** what outcome would refute the construction or its stronger interpretation?

Stop rule:

A combination without a typed operation does not generate an object. Verbal or visual resemblance alone is marked [Refuted].

The complete result for all 305 combinations is stored in `FIN_Second_Generation_Combination_Search.csv`. Each row records the common operation, candidate object, object space, new observable, new dynamics, status, and whether the result is a new coupling or merely inherits a lower-order object.

Chapter 3

Base inventory C01...C12

ID	Base object	Dominant operation	Critical limitation
C01	strict kernel	convolution/spectralization on Z_{12}	no automatic continuum
C02	legacy intermediate bridge	propagator-like damped profile	no complete completion map or role transfer
C03	$A = sI - W \succeq 0$	Dirichlet form and spectral calculus	no energy unit
C04	e^{-itA}, e^{-tA}	two rays of functional calculus	different operational semantics
C05	Green/resolvent/action	operator inversion and variation	propagator does not determine the full interaction
C06	cocycle/chiral ceivers	re-sign under inversion	no non-premise sign
C07	fractal Schur compression	elimination of degrees of freedom	strict is not a static fixed point
C08	adaptation and memory	operator-record feedback	learning law is additional input
C09	information/geometry	entropy, Fisher metric, diffusion	no physical unit
C10	operational process	preparation-instrument-environment-record	concrete laboratory remains external
C11	scale obstruction	quotient by $\mathbb{R}_{>0}$	no canonical section
C12	selector obstruction	orientation quotient/torsor	QW-2191 remains open

Chapter 4

Main table: second-generation objects

FIN combination	Derived object	Mathematical form	Analogous fields	New intuition	Status
C03+C05+C07	O01 Dynamic Spectral Compression Kernel	$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}$	Feshbach, Zwanzig, reduction	Mori-Kron dynamic reduction generates memory	[Proven]
C03+C05+C07	O02 Frequency-Dependent Effective Action	$S_z[x] = \frac{1}{2}\langle x, (zI + A_{EE} - \Sigma_E(z))x \rangle - \langle j, x \rangle$	effective action, influence functional	Green, compression, and variation become one object	[Proven]
C05+C07+C10	O03 Resolvent text Presheaf	$E \mapsto \text{Schur}_{V \setminus E}(zI + A)$	category theory, open systems, network reduction	each observer/context has a compatible effective operator	[Proven]
C03+C06+C12	O04 Chiral Spectral Response Bundle	$(M_R, \Lambda, C_\chi, \text{Im } B)$	asymmetry resources, harmonic analysis	distinct sign receivers detect one resource type	[Strong evidence]
C06+C07+C10	O05 Chiral Memory Susceptibility	$\Xi_E(z) = \partial_\theta \Sigma_E(z, \theta) _0$	Kubo response, flux susceptibility, open systems	hidden degrees of freedom possess chiral memory response	[Strong evidence]
C03+C04+C09	O06 Heat Information Metric Tower	$D_t^2(x, y) = \sum_z P_t(x, z) - P_t(y, z) ^2 / \pi_z$	diffusion maps, information geometry	informational distance is multiscale	[Proven]
C10+C11	O07 Operational Calibration Torsor	$(A, \tau) \sim (cA, \tau/c)$	metrology, gauge fixing, identification	experiment selects scale; algebra supplies an orbit	[Proven]
C01+C03+C11	O08 Projective Spectral Fingerprint	$(\lambda_1/\lambda_{\max}, \dots)$	inverse spectral theory, control	part of the prediction is testable before absolute calibration	[Proven]
C04+C09+C10	O09 Information Contraction Ledger	$\mathcal{L}_t = D(p\ q) - D(P_t p\ P_t q)$	data processing, statistics, sensory bottlenecks	"loss" is loss of accessible distinguishability	[Proven]
C04+C07+C10	O10 Dual-Ray Reduced Process Tensor	$\mathcal{T}_r[\mathcal{M}_1, \dots, \mathcal{M}_r]$ for wave/heat projections	process tensors, cybernetics, models	memory becomes visible through multitime causal interventions	[Moderate evidence]
C08+C09+C10	O11 Adaptive Memory Geometry Flow	$(\dot{A}, \dot{q}) = (-\nabla_A \mathcal{L}, -\text{grad}_F \mathcal{L})$	neural operators, adaptive networks, plasticity	learning may occur in memory-geometry space	[Speculative]
C05+C07+C08	O12 Multiscale Memory Functor	$\Sigma^{(0)}(z) \rightarrow \Sigma^{(1)}(z) \rightarrow \dots$	continued fractions, hierarchical reservoirs, RG	seek a fractal fixed point in memory-function space	[Strong evidence]
C01+C02+C07	O13 Legacy-Strict Completion-Defect Tower	$\Delta_n = K_{\text{strict}}^{(n)} - \mathcal{C}_n[K_{\text{legacy}}^{(n)}]$	error propagation, RG defects, obstruction theory	study bridge failure dynamically instead of asserting similarity	[Moderate evidence]
C06+C09+C12	O14 Spectral Asymmetry Transport Cost	$\inf_{\Phi \text{ covariant}} \text{Cost}(\Phi)$	optimal transport, resource conversion	orientation has operational cost, but cost does not select a sign	[Moderate evidence]
C10+C11+C12	O15 Two-Torsor Physicalization Bundle	$\mathcal{T}_{\text{scale}} \times \mathcal{T}_{\text{orient}}$	principal bundles, frames, calibration	scale and orientation require separate sections	[Strong evidence]

Chapter 5

Explanatory power and falsification

Derived object	What it explains	What it does NOT explain	Falsifying test
O01 DSCK	exact memory after elimination	physical environment and scale	constant $\Sigma_E(z)$ despite nonzero coupling
O02 Effective Action	exact contextual source response	nonlinear interactions and quantization	stationary solution disagrees with block Green function
O03 Context Presheaf	compatibility of nested reductions	unique observer context	nonzero direct-versus-nested residual
O04 Chiral Bundle	common covariance of sign receivers	source of sign	symmetric state with a nonzero odd receiver
O05 Chiral Memory	memory response to phase twist	selector for θ	loss of oddness under reflection
O06 Metric Tower	geometry of distinguishability	SI length or Lorentz signature	distance grows under a contractive semigroup
O07 Calibration Torsor	non-identifiability of absolute scale	laboratory reference source	distinguish (A, τ) from $(cA, \tau/c)$ using the same P_τ
O08 Fingerprint	scale-free spectral predictions	absolute energy	fingerprint changes under positive rescaling
O09 Information Ledger	accessible loss of distinguishability	thermodynamic energy or ontological destruction	violation of data processing
O10 Process Tensor	operational memory and interventions	concrete microscopic dilation	a memoryless model reproduces all interventions
O11 Adaptive Flow	possible operator-learning mechanism	strict source law or biology	no hold-out advantage over a static model
O12 Memory Functor	hierarchical dynamic compression	a strict fixed point	no stabilization of any functional class
O13 Defect Tower	where the bridge fails	role transfer and completion theorem	residual converges to zero without target coding
O14 Transport Cost	amount of asymmetry resource required	canonical choice of that resource	free operation increases the monotone
O15 Two-Torsor Bundle	independence of scale and orientation problems	source of either section	one internal datum canonically closes both torsors

Chapter 6

Ranking O01...O15

Scores use novelty, mathematical rigor, FIN relevance, and falsifiability from 0 to 5. Overinterpretation risk also ranges from 0 to 5, where 5 is highest.

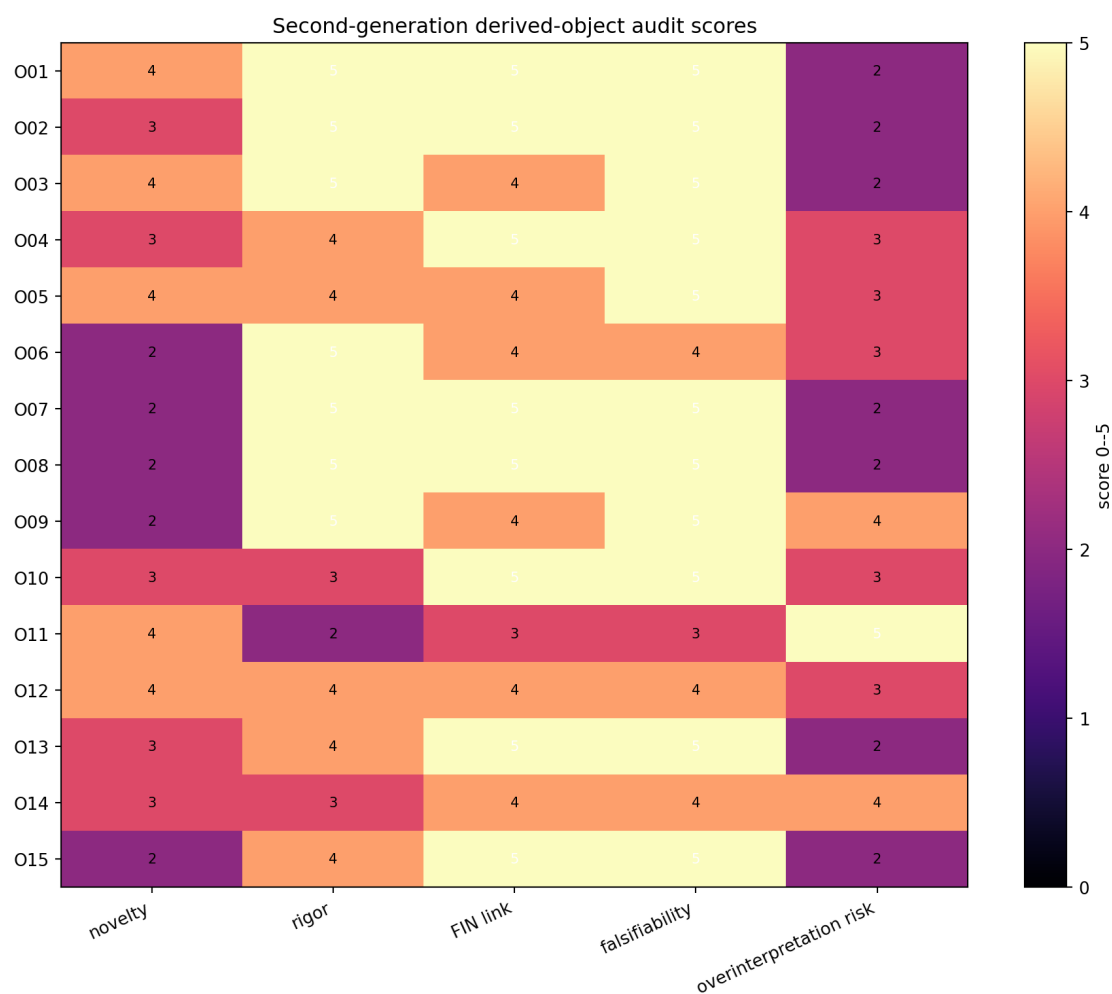


Figure 6.1: Audit scores for O01–O15. Overinterpretation risk has the opposite semantics from the other scores.

ID	Novelty	Rigor	FIN	Falsifiability	Risk
O01	4	5	5	5	2

ID	Novelty	Rigor	FIN	Falsifiability	Risk
O02	3	5	5	5	2
O03	4	5	4	5	2
O04	3	4	5	5	3
O05	4	4	4	5	3
O06	2	5	4	4	3
O07	2	5	5	5	2
O08	2	5	5	5	2
O09	2	5	4	5	4
O10	3	3	5	5	3
O11	4	2	3	3	5
O12	4	4	4	4	3
O13	3	4	5	5	2
O14	3	3	4	4	4
O15	2	4	5	5	2

O01 has the best balance. O01+O03+O12 yields the richest higher-order structure. O11 has the greatest overinterpretation risk because analogies with learning and biology are not FIN source laws.

Chapter 7

Theorems generated in the second phase

7.1 O01 belongs to the operator Stieltjes class

For $z > 0$ and $A_{HH} \succeq 0$,

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}.$$

For every $n \geq 0$,

$$(-1)^n \Sigma_E^{(n)}(z) = n! A_{EH}(zI + A_{HH})^{-(n+1)} A_{HE} \succeq 0.$$

Thus Σ_E is a completely monotone positive operator-valued function. **[Proven]**

Its poles and spectral measure encode hidden modes; its inverse Laplace transform is a decaying memory kernel. Consequently, the admissible space for dynamic compression is far narrower than the set of all matrix-valued functions. For strict Z_{12} , orders $0, \dots, 4$ have minimum eigenvalues above -6.4×10^{-17} , consistent with positivity up to floating-point error.

7.2 O03 composes functorially

Let $G \subset F \subset E \subset V$. For $B(z) = zI + A$,

$$\text{Schur}_{E \setminus G} \left(\text{Schur}_{V \setminus E} B \right) = \text{Schur}_{V \setminus G} B.$$

This is associativity of Gaussian elimination/the Schur complement. **[Proven]** The direct-versus-nested residual in selected strict Z_{12} contexts is 2.47×10^{-16} .

"Functor" is used in the restricted sense that the context poset and reduction morphisms form a compatible operator diagram. No sheaf axioms or quantum contextuality theorem are claimed.

7.3 O02 reconstructs the exact contextual Green function

Set

$$B_E(z) = zI + A_{EE} - \Sigma_E(z).$$

The functional

$$S_z[x; j] = \frac{1}{2} \langle x, B_E(z)x \rangle - \langle j, x \rangle$$

has stationary solution

$$x_\star = B_E(z)^{-1}j = \left[(zI + A)^{-1} \right]_{EE} j.$$

The audited stationarity residual is 4.90×10^{-16} . **[Proven]** This exactly reconstructs an effective action from the resolvent. It is not a complete physical action because z , the action unit, measure, and interactions remain additional data.

7.4 O05: memory has a chiral response direction

Introduce a Hermitian twisted family $A(\theta)$, in which a hop by d acquires phase $e^{id\theta}$. Then

$$\Xi_E(z) = \partial_\theta \left[A_{EH}(\theta)(zI + A_{HH}(\theta))^{-1} A_{HE}(\theta) \right] \Big|_{\theta=0}.$$

For the even/odd split of strict Z_{12} ,

$$\|\Xi_E(0.2)\|_2 = 1.2175076427.$$

For reflection R ,

$$R_E \Xi_E(z) R_E = -\Xi_E(z),$$

with residual 2.17×10^{-16} . **[Proven]** for the stated twisted family and split.

Likewise,

$$C_\chi = \frac{d}{d\theta} \text{Tr}[\rho A(\theta)] \Big|_{\theta=0}$$

has residual 3.23×10^{-13} , promoting C_χ from an ad hoc indicator to an explicit flux-twist response. The twist and its sign are still inserted in the test family; O05 does not discharge QW-2191.

7.5 O09: negative informational coupling as channel contraction

For a Markov channel P_t and positive distributions p, q ,

$$D(P_t p \| P_t q) \leq D(p \| q).$$

Define

$$\mathcal{L}_t(p, q) = D(p \| q) - D(P_t p \| P_t q) \geq 0.$$

Across 1000 deterministically reproducible pairs:

Quantity	Result
minimum $\mathcal{L}_t$	0.0485457
median	0.718427
maximum	3.15526
violations below -10^{-12}	0

This is **[Proven]** as an instance of the data-processing theorem. It is methodologically preferable to inserting "negative information" into a scalar kernel because it identifies the two hypotheses, the contracting channel, and the difference between inaccessible and destroyed information. It does not pretend to be energy without temperature and scale.

7.6 O07+O08: exact calibration orbit

$$e^{-\tau A} = e^{-(\tau/c)(cA)}.$$

The residual at $c = 7.25$ is 3.43×10^{-17} , and the spectral fingerprint remains invariant with residual 6.66×10^{-16} . **[Proven]**

An experiment can therefore test a scale-free fingerprint first, but physical time requires an external calibration section.

7.7 O06: heat geometry contracts

The maximum squared diffusion distance on Z_{12} is:

t	$\max_{x,y} D_t^2(x, y)$
0.10	17.3358
0.25	11.0222
0.50	5.69186
1.00	2.00931

[Proven] for the tested family. The information geometry exists, but it is dimensionless and semigroup-time dependent.

Chapter 8

Second-order analogies

FIN ject	ob- First analogy	Derived phenomenon	Returning FIN intuition	Boundary
O01	Feshbach/Mori–Zwanzig	self-energy, generalized Langevin equation	hidden nodes generate memory	split is supplied
O02	effective action	influence functional	contextual resolvent has a variational source	no complete QFT
O03	context categories	compatible restrictions and diagrams	observer as context selection	no contextuality theorem
O04	asymmetry modes	resource monotones, frames	receivers can be classified harmonically	no source resource
O05	Kubo/flux response	susceptibility, pumping	memory has a chiral tangent	no topological quantization
O06	diffusion maps	data geometry, coarse graining	heat generates internal geometry	no spacetime
O07	gauge fixing/metrology	scale section	calibration belongs to operational theory	not a strict source
O08	inverse spectra	system identification	test ratios before units	false positives possible
O09	data processing	sensory bottleneck, sufficiency	information can enter hidden correlations	no Landauer energy
O10	process tensors	non-Markovian interventions	multitime records distinguish memory	full instrument required
O11	neural operators	maps between function spaces	learn $\Sigma(z)$, not only a matrix	learning is not ontology
O12	hierarchical reservoirs	fading memory, continued fractions	fixed point may live in Stieltjes space	theorem remains open
O13	RG defect propagation	relevant/irrelevant directions	bridge defect may have a flow exponent	target coding forbidden
O14	optimal transport	minimum conversion cost	orientation is an operational resource	cost does not choose sign
O15	principal bundles	calibration and frames	two independent missing objects	sections remain external

8.1 Physics

The strongest mechanical analogies are Feshbach projection/self-energy, Mori–Zwanzig memory kernels, open-system process tensors, linear response to a twist or flux, and effective actions after field elimination. There is no equivalence to a concrete field theory because FIN still lacks local spacetime, fields, a measure, an action scale, and quantization rules.

8.2 Mathematics

The natural new space is

$$\text{Stieltjes}_+(E) = \left\{ \Sigma : (0, \infty) \rightarrow \text{End}(E) \mid (-1)^n \Sigma^{(n)}(z) \succeq 0 \right\}.$$

Nested Schur complements form a diagram over the context poset. Sheaf/global section language remains a research analogy until covering data and gluing axioms are defined.

8.3 Computer science

O11 resembles neural operators because the learned object is the map

$$\text{process record} \mapsto \Sigma(z),$$

not a single parameter vector. O12 resembles reservoir computing: the hidden sector retains a trace of history and the current record is a readout. This does not establish a biological or computational ontology for the nadsoliton.

8.4 Biology

The defensible analogy concerns fading memory and adaptive weights: hidden modes represent decay scales, plasticity would modify the memory spectrum, and readout sees only a projection of the full state. **[Speculative]** No biological data or cell/synapse-to-FIN-node map exists.

8.5 Cybernetics

O03, O07, O08, and O10 form the scheme

$$\text{system} \rightarrow \text{observer} \rightarrow \text{identification} \rightarrow \text{calibration} \rightarrow \text{decision}.$$

The missing observer need not be a new term in A ; it may be a family of projections, instruments, and calibration sections.

Chapter 9

Third generation: combining O-objects

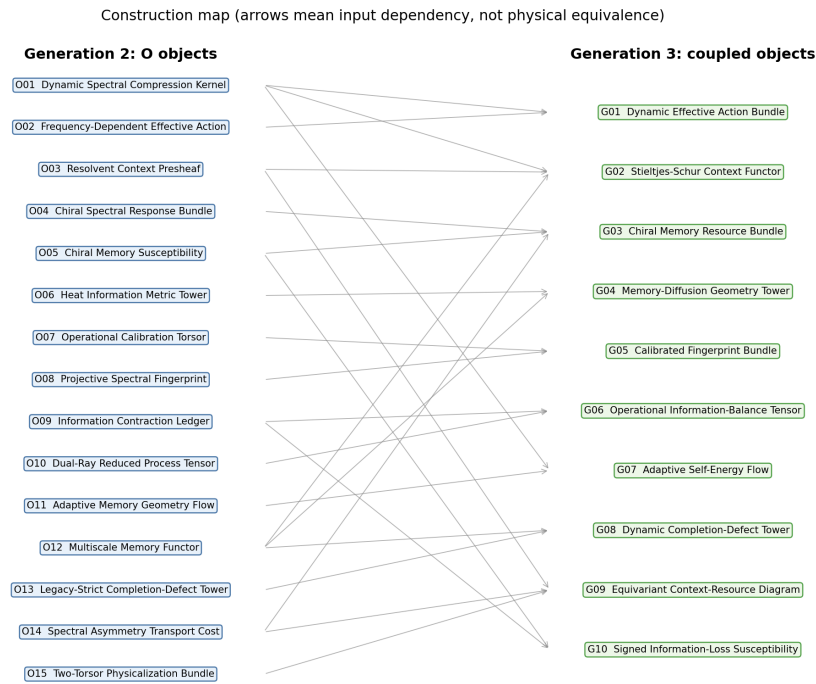


Figure 9.1: Input dependencies between the O and G generations. Arrows do not assert physical equivalence.

ID	Combination	Higher-order object	Definition/role	Status
G01	O01+O02	Dynamic Effective Action Bundle	Stieltjes self-energy plus exact source action	[Proven]
G02	O01+O03+O12	Stieltjes-Schur Context Functor	contexts, compatible reductions, completely monotone self-energies	[Proven]
G03	O04+O05+O14	Chiral Memory Resource Bundle	odd memory susceptibility plus asymmetry-resource budget	[Strong evidence]
G04	O06+O12	Memory-Diffusion Geometry Tower	geometry indexed by diffusion time and hiding depth	[Moderate evidence]
G05	O07+O08	Calibrated Fingerprint Bundle	scale-free test plus explicitly external scale section	[Proven]
G06	O09+O10	Operational Information-Balance Tensor	multitime information-contraction ledger	[Moderate evidence]
G07	O01+O11	Adaptive Self-Energy Flow	learned trajectory in Stieltjes-function space	[Speculative]
G08	O12+O13	Dynamic Completion-Defect Tower	tests whether bridge defect survives dynamic reduction	[Moderate evidence]
G09	O03+O14+O15	Equivariant Context-Resource Diagram	contexts, scale, orientation as separate sectional structures	[Speculative]

ID	Combination	Higher-order object	Definition/role	Status
G10	O05+O09	Signed Information-Loss Susceptibility	derivative of information contraction with respect to chiral twist	[Speculative]

Chapter 10

Principal "shadow shapes"

10.1 Shadow 1 — the missing object space is Stieltjes space

Rather than ask which next static kernel arises after compression, the Green+Schur+dynamics combination shows that the proper object is the function $z \mapsto \Sigma(z)$. The self-similarity question changes from "is K a fixed point?" to "is the family of memory functions closed or attracted to a fixed class?" [\[Strong evidence\]](#) as a direction; no fixed point was found.

10.2 Shadow 2 — observation and compression both select context

Compression selects retained degrees of freedom; an observer selects accessible observables. Their common shadow is the context E and its effective operator $B_E(z)$. [\[Moderate evidence\]](#) Physical observation is not thereby reduced to a Schur complement: an instrument and record remain necessary.

10.3 Shadow 3 — "negative information" is often flow into an invisible sector

O09 and O01 suggest

loss of accessible distinguishability $\longleftrightarrow$ correlations/excursions in sector H .

Information may remain in a full reversible description while disappearing from the selected record. [\[Strong evidence\]](#)

10.4 Shadow 4 — chirality is a tangent, not a point

$$\Xi_E \in T_{\Sigma_E} \text{Stieltjes}_+(E).$$

Orientation appears as a direction in memory-operator space. This new derived characteristic does not choose whether that tangent is positive or negative. [\[Strong evidence\]](#)

10.5 Shadow 5 — physicalization requires two sections and one process

The smallest shadow of missing physics consists of three independent elements: a scale section, an orientation/sector section, and an operational process with preparation, instrument, and record. None substitutes for either of the others. **[Proven]** as a separation of types.

10.6 Shadow 6 — adaptation may act on memory rather than directly on a kernel

Instead of

$$\dot{K} = -\nabla_K \mathcal{L},$$

the natural object may be

$$\partial_t \Sigma(z; t) = -\text{grad}_\Sigma \mathcal{L}_{\text{process}}.$$

[Speculative] Positivity and complete monotonicity could constrain the learning law.

10.7 Shadow 7 — the bridge defect may be dynamic

Legacy and strict may differ not only by their distance profile but by hidden memory structure. O13/G08 can test whether the defect decays, remains constant, grows, or leaves the Stieltjes class. None of these outcomes is a bridge by assumption; it is a new falsifiable atom.

Chapter 11

Falsification of the strongest interpretations

11.1 Is G02 a new category theory?

No. **[Refuted]** Schur associativity and context diagrams are known mathematics. Potential novelty lies in using them to organize FIN.

11.2 Does O05 solve the selector problem?

No. **[Refuted]** The family $A(\theta)$ requires θ , and inversion maps θ to $-\theta$, producing the pair $\pm\Xi$. O05 is a new receiver.

11.3 Is O09 physical entropy?

Not without additional data. **[Refuted]** Thermodynamics requires at least a temperature, Hamiltonian, protocol, and an apparatus/reset ledger.

11.4 Does O12 prove a fractal fixed point?

No. **[Refuted]** O12 defines the right search space. Existence of an attractor or self-similar operator measure remains open.

11.5 Does O11 prove that FIN is biological?

No. **[Refuted]** Neural operators, reservoir computing, and plasticity provide mechanistic analogies, but no empirical map.

11.6 Does G09 provide a global section?

No on current artifacts. **[Refuted]** The sectional language exposes the obstruction; it removes neither QW-2191 nor the scale orbit.

Chapter 12

Research programs 255–266

1. **P255 — Stieltjes theorem for all contexts.** Prove positivity, complete monotonicity, operator-measure representation, and the limits at zero and infinity. Probability: 0.95.
2. **P256 — Minimal realization of self-energy.** Classify hidden pairs (A_{HH}, A_{HE}) producing the same $\Sigma_E(z)$. Probability: 0.85.
3. **P257 — Formalize the context functor.** Define the category and verify Schur associativity in a proof assistant or independent algebraic core. Probability: 0.80.
4. **P258 — Analytic chiral memory susceptibility.** Derive $\Xi_E(z)$, covariance, and norm bounds while separating gauge convention from flux observable. Probability: 0.82.
5. **P259 — Dynamic RG in Stieltjes space.** Define $\mathcal{R} : \Sigma^{(n)} \mapsto \Sigma^{(n+1)}$ and search for fixed points, cycles, or no-go results. Probability: 0.45; high reward.
6. **P260 — Multitime information-balance tensor.** Extend O09 to a process ledger separating system, apparatus, and hidden-node contraction. Probability: 0.75.
7. **P261 — Dynamic completion-defect test.** Freeze one map and test $\Delta_\Sigma(z) = \Sigma_{\text{strict}}(z) - \mathcal{C}_\Sigma[\Sigma_{\text{legacy}}(z)]$, without fitting after seeing the target. Bridge probability: 0.30; valuable no-go: 0.80.
8. **P262 — Calibrated Fingerprint Experiment.** Test spectral ratios first, then independently calibrate τ , with no post-unblinding changes. Methodological success: 0.75.
9. **P263 — Tomography of currents and chiral memory.** Measure C_d , $d = 1, \dots, 5$, and intermediate interventions; test C_χ and $\Xi_E(z)$ separately. Probability: 0.70.
10. **P264 — False-positive atlas O01–O10.** Quantify how often random and structured Laplacians pass the FIN tests. Probability: 0.90.
11. **P265 — Identifiability of Adaptive Memory Geometry.** Compare genuine generator drift with static operators, apparatus memory, and calibration errors using hold-out data. Probability: 0.65.
12. **P266 — Biological/cybernetic benchmark.** Test fading-memory capacity, robustness–plasticity, recovery, and readout dimension against matched standard reservoirs. Probability: 0.72.

Chapter 13

Strategic priority

Rank	Program	Reason
1	P255	near-certain formal closure of the new object space
2	P256	determines what can be inferred about the hidden sector
3	P258	closes the new chiral-memory characteristic without selector claim
4	P257	protects context language against metaphorical use
5	P264	measures specificity of the atlas
6	P260	joins information to a complete process
7	P262	moves the result toward the laboratory
8	P263	separates harmonics from memory experimentally
9	P259	highest mathematical reward, high risk
10	P261	one honest bridge attack
11	P265	tests adaptation and identifiability
12	P266	controls biological/cybernetic analogies

Chapter 14

Guardrail-compatible boundaries

This report does not export a non-premise strict selector, discharge QW-2191, a canonical unit of length/time/mass/energy/action, a complete legacy-to-strict map, role transfer for $\sin^2 \theta_W$, α_{EM} , or the gravity hierarchy, a strict-source adaptive law, L_{total} , the Standard Model, GR, a ToE, a physical observer, environment or apparatus, or external data validation.

The nadsoliton remains primordial information in a solitonic state. The contexts E/H partition accessibility of its degrees of freedom; they are not a separate informational layer beneath it.

Chapter 15

Final verdict

The most important derived structure is neither another kernel nor another constant. It is

the Stieltjes–Schur Context Functor

,

a contextual family of exact effective operators whose self-energies are completely monotone, whose reductions compose, and whose time transforms generate memory.

It organizes the common shadow

Green $\leftrightarrow$ compression $\leftrightarrow$ memory $\leftrightarrow$ observer $\leftrightarrow$ accessible information.

The most interesting new tangent characteristic is $\Xi_E(z)$, the chiral memory susceptibility. The most defensible account of information loss is $\mathcal{L}_t$, contraction of accessible distinguishability. The central no-go remains the absence of orientation and scale sections.

This does not establish FIN as physics. It does provide a rigorous new object space in which subsequent questions are better typed and easier to falsify.

Chapter 16

Comparative literature

1. Dörfler, F.; Bullo, F., *Kron Reduction of Graphs with Applications to Electrical Networks*, <https://arxiv.org/abs/1102.2950>.
2. Lin, Y. T.; Tian, Y.; Anghel, M.; Livescu, D., *Data-driven learning for the Mori–Zwanzig formalism*, <https://arxiv.org/abs/2101.05873>.
3. Pollock, F. A. et al., *Non-Markovian quantum processes: complete framework and efficient characterisation*, <https://arxiv.org/abs/1512.00589>.
4. Marvian, I.; Spekkens, R. W., *Modes of asymmetry*, <https://arxiv.org/abs/1312.0680>.
5. Coifman, R. R.; Lafon, S., *Diffusion maps*, <https://doi.org/10.1016/j.acha.2006.04.006>.
6. Shuman, D. I. et al., *The Emerging Field of Signal Processing on Graphs*, <https://arxiv.org/abs/1211.0053>.
7. Kovachki, N. et al., *Neural Operator: Learning Maps Between Function Spaces*, <https://arxiv.org/abs/2108.08481>.
8. Abramsky, S.; Brandenburger, A., *The Sheaf-Theoretic Structure of Non-Locality and Contextuality*, <https://arxiv.org/abs/1102.0264>.
9. Chamseddine, A. H.; Connes, A., *The Spectral Action Principle*, <https://arxiv.org/abs/hep-th/9606001>.

Chapter 17

Reproducible artifacts

- `fin_nadsoliton_second_generation_atlas.py`
- `FIN_Second_Generation_Combination_Search.csv`
- `FIN_Second_Generation_Derived_Objects.csv`
- `FIN_Second_Generation_Generation3.csv`
- `FIN_Second_Generation_Atlas_Results.json`
- `FIN_Second_Generation_Atlas_Figures/combination_search_coverage.png`
- `FIN_Second_Generation_Atlas_Figures/derived_object_score_matrix.png`
- `FIN_Second_Generation_Atlas_Figures/generation2_generation3_map.png`
- `FIN_Second_Generation_Atlas_Derived_Nadsoliton_Objects_EN.md`

Reproduction certificate

```
MPLCONFIGDIR=/tmp/matplotlib-second \  
python3 fin_nadsoliton_second_generation_atlas.py
```

Expected coverage: 305 combinations, 15 O-generation objects, and 10 G-generation objects. The construction grammar is a candidate filter, not proof of equivalence or physical realization.